

Question	Answer	Marks
4(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): acceleration is (always) in the opposite <u>direction</u> to displacement or acceleration is (always) <u>directed</u> towards a fixed point	B1
4(b)(i)	amplitude = $(9.5 - 3.5) / 2$ = 3.0 cm	A1
4(b)(ii)	$\omega = v_0 / x_0$	C1
	= $9.5 / 3.0 = 3.2 \text{ rad s}^{-1}$	A1
4(b)(iii)	$T = 2\pi / \omega$	C1
	= $2\pi / 3.2$ = 2.0 s	A1
4(b)(iv)	attempted sinusoidal curve starting with a minimum at $t = 0$	B1
	sinusoidal curve of period 2.0 s from $t = 0$ to $t = 6.0$ s	B1
	all peaks shown at $h = 9.5$ cm	B1
	all troughs shown at $h = 3.5$ cm	B1

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